11.9.4 Comparison between soap and detergents

Soap	Detergent
It is a sodium salt of long chain fatty acids.	It is sodium salts of sulphonic acids.
The ionic part of a soap is $-COO^-Na^+$.	The ionic part in a detergent is $-SO^-_3Na^+$.
It is prepared from animal fats or vegetable oils.	It is prepared from hydrocarbons obtained from crude oil.
Its effectiveness is reduced when used in hard water.	It is effective even in hard water.
It forms a scum in hard water.	Does not form a scum in hard water.
It has poor foaming capacity.	It has rich foaming capacity.
Soaps are biodegradable.	Most of the detergents are non-biodegradable.

Have you noticed the term "TFM" in soap



TFM means TOTAL FATTY MATTER. It is the one of the important factors to be considered to assess the quality of soap. A soap, which has higher TFM, is a good bathing soap.

Points to Remember

- A group or class of organic compounds related to each other by a general molecular formula constitutes homologous series.
- The IUPAC name of the any organic compound consist of three parts. **ROOTWORD, PREFIX and / or SUFFIX**.
- Functional group may be defined as an atom or group of atom or reactive part which is responsible for the characteristic properties of the compounds
- Ethanoic acid is most commonly known as acetic acid and belongs to a group of acids called carboxylic acids.
- Ethanol or ethyl alcohol or simply alcohol is one of the most important members of the family of alcohols.
- The slow chemical change that takes place in complex organic compounds by the action of enzymes leading to the formation of simple molecules is called fermentation.
- Soaps are sodium or potassium salts of some long chain carboxylic acids.
- Detergents are sodium salts of sulphonic acids. Thus instead of –COOH group in soaps, detergents contain $-SO_3H$ group

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